

Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

2. Consider 4 employers $\{e_1, e_2, e_3, e_4\}$ and 4 applicants $\{a_1, a_2, a_3, a_4\}$.

preferences for employers					preferences for applicants						
e_1	:	2	3	4	1	a_1	:	3	2	1	4
e_2	:	3	2	1	4	a_2	:	3	2	1	4
e_3	:	3	2	1	4	a_3	:	1	3	4	2
e_4	:	3	2	4	1	a_4	:	3	4	2	1

Preferences are listed from high to low for employers and applicants.

- Run the Gale-Shapley algorithm on the above input. Show all stages of the algorithm.
- Describe what happens when the Gale-Shapley algorithm needs many rounds of offers to terminate. In particular, for which type of inputs does this happen?

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Answer:

2. (a)
 - i. Employer 1 offers to applicant 2, $S = \{(1, 2)\}$.
 - ii. Employer 2 offers to applicant 3, $S = \{(1, 2), (2, 3)\}$.
 - iii. Employer 3 offers to applicant 3, applicant 3 breaks with employer 2, accepts offer, $S = \{(1, 2), (3, 3)\}$.
 - iv. Employer 2 offers to 2; applicant 2 breaks with employer 1, accepts offer, $S = \{(2, 2), (3, 3)\}$.
 - v. Employer 1 offers to applicant 3, applicant 3 breaks with employer 3, accepts offer, $S = \{(2, 2), (1, 3)\}$.
 - vi. Employer 3 offers to applicant 2, applicant 2 breaks with employer 2, accepts offer, $S = \{(3, 2), (1, 3)\}$.
 - vii. Employer 2 offers to 1, $S = \{(3, 2), (1, 3), (2, 1)\}$.
 - viii. Employer 4 offers to applicant 3, who rejects, because employer 4 is lower ranked than employer 1.
 - ix. Employer 4 offers to applicant 2, who rejects, because employer 4 is lower ranked than employer 3.
 - x. Employer 4 offers to applicant 4, $S = \{(3, 2), (1, 3), (2, 1), (4, 4)\}$.
- (b) In the example, 10 offers are made to 4 applicants. This happens when lower ranked employer make early offers. When higher ranked employer make offers later, then applicants reject the lower ranked employers who made early offers.

Looking at the input matrices, we observe that employers prefer the applicants 2 and 3 over applicants 1 and 4, whereas employer 3 is highly ranked among the applicants. Employer 3 start offering after employers 1 and 2 who are lower in the preferences of the applicants.